

Chapter 03

1. What is a program?

Program is a sequence of steps. For each step, an arithmetic or logic operation is done. For each operation, a different set of control signal is needed.

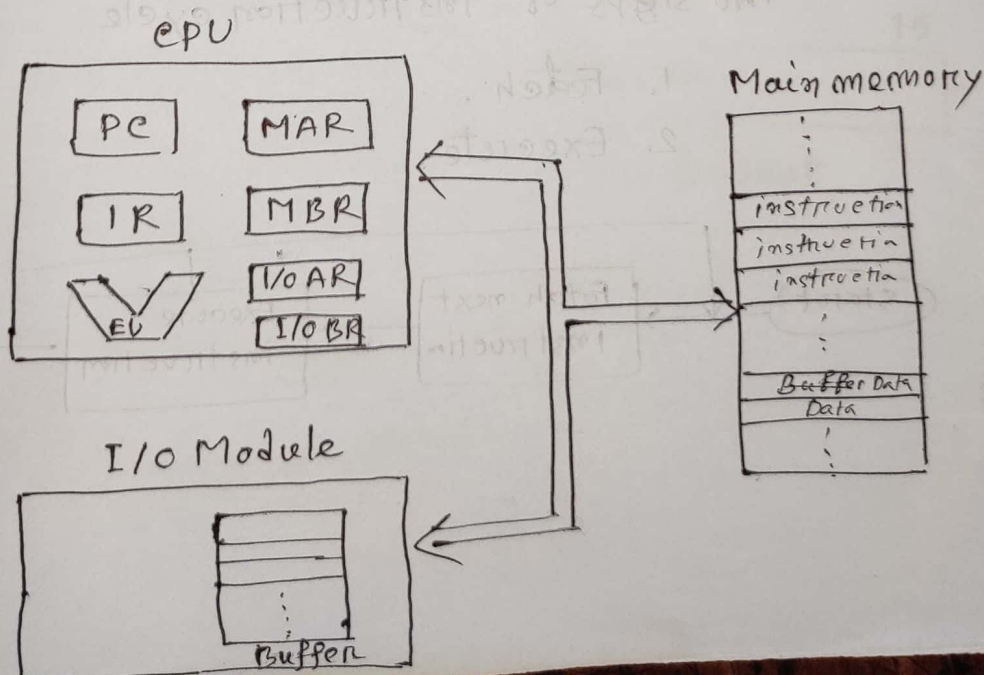
2. Computer Components

1. Central processing unit

- * Control unit
- * Arithmetic logic unit

2. Input/output

3. Main memory



Program Counter $\rightarrow$ PC

Instruction register $\rightarrow$ IR

Memory address register $\rightarrow$ MAR

Memory buffer register $\rightarrow$ MBR

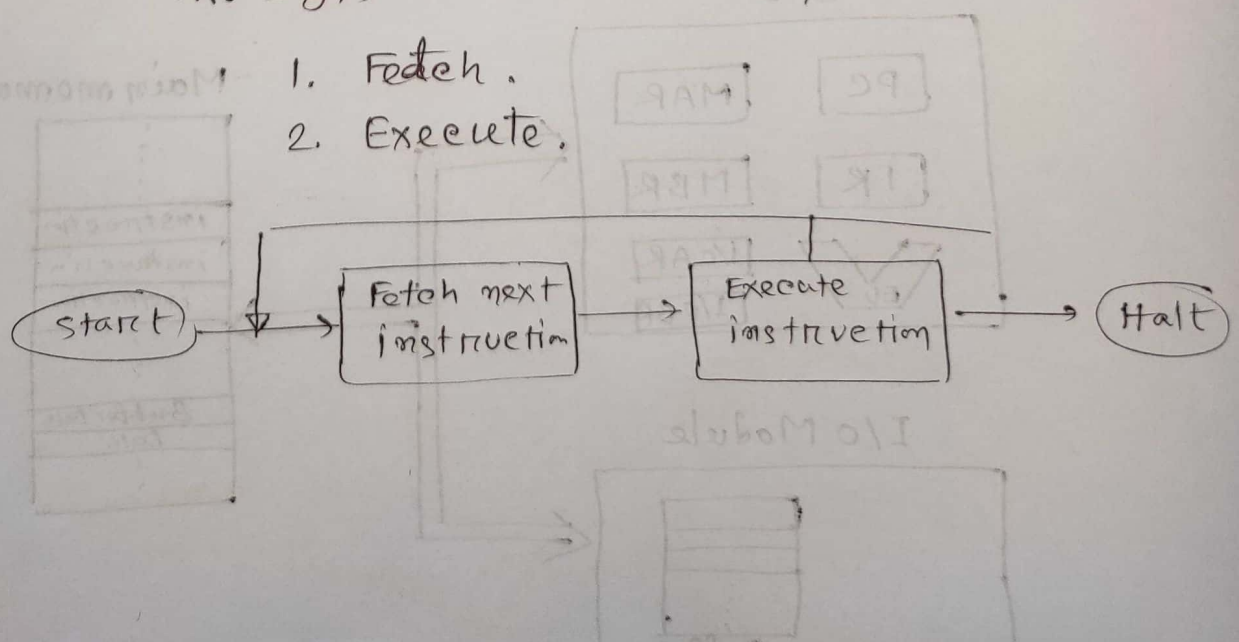
Input output address register $\rightarrow$ O/I AR

Input output buffer register $\rightarrow$ I/O BR

3. Instruction Cycle:

Computer function: The basic function performed by the computer is execution of a program, which consists of a set of instructions stored in a memory.

Two steps of instruction cycle

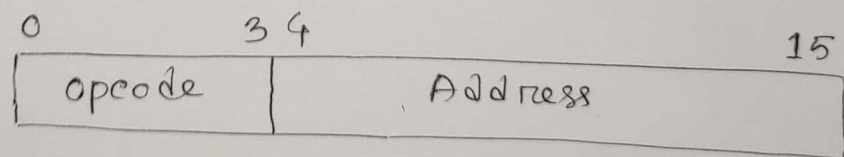


Fetch cycle.

1. pc
2. Processor
3. Increment pc
4. IR
5. Processor interprets

Execute cycle :

1. Processor - memory
2. Processor I/O
3. Data processing
4. Control



instruction format

Example of program Execution

Instruction Set.

0001 → load Ac from M
0010 → store Ac to M
0101 → Add Ac with M

Problem: Do addition two memory content by using Ac. and store into memory.

N.B: memory location 95

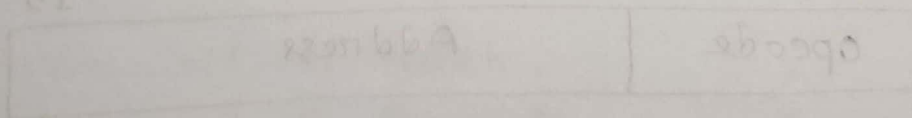
Step 1 : load

Step 2 : Addition

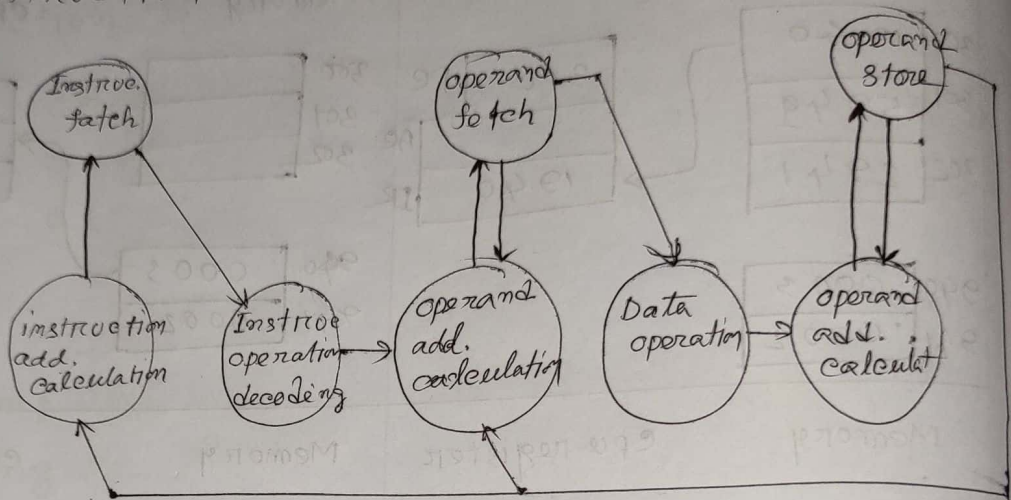
Step 3 : store in memory

First operand location 940

Add 0003
0002



Instruction Cycle Diagram

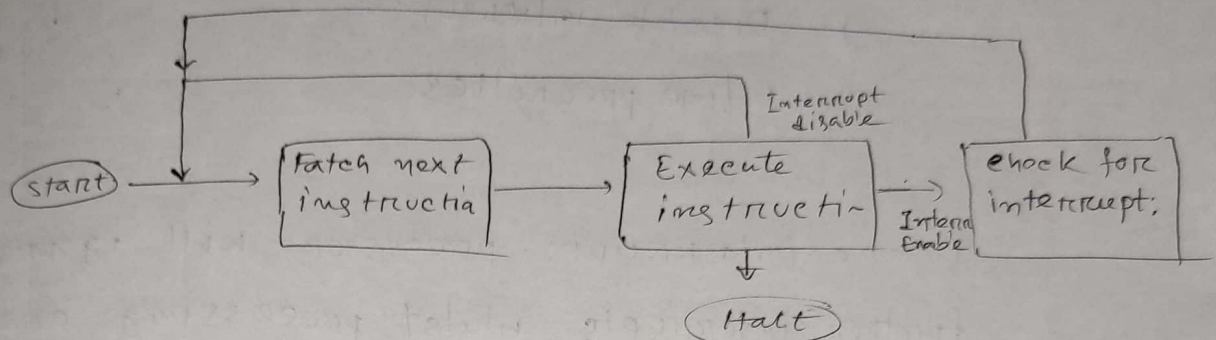


Interrupt: An interrupt is a condition that halt the microprocessor temporarily to work on a different task and then return to its previous task.

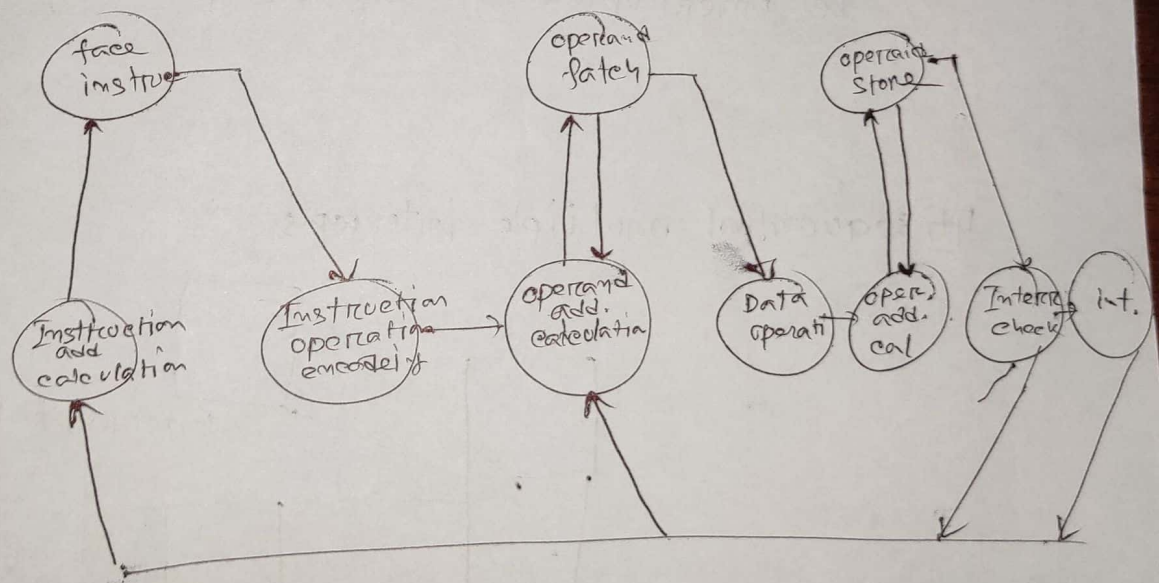
classes of interrupts:

1. Program :
2. Timer :
3. I/O :
4. Hardware failure :

Instruction Cycle with interrupts



Instruction cycle (with interrupts) state diagram



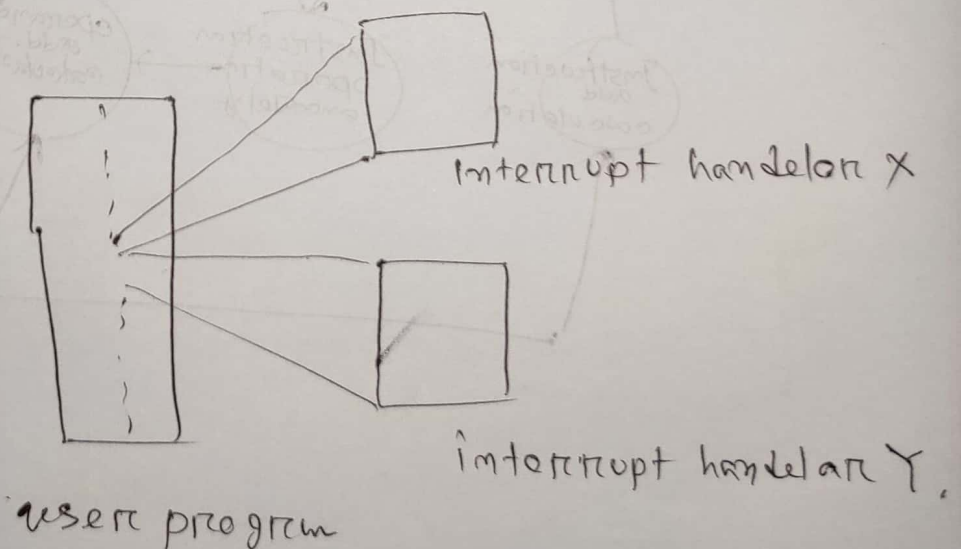
Multiple Interrupts:

1. Disable interrupts
2. Define priorities

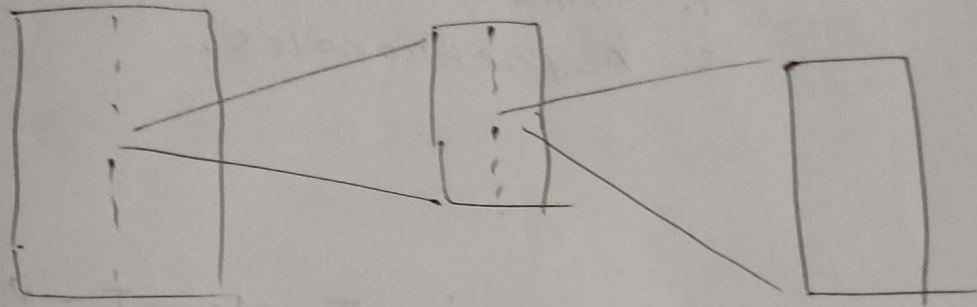
Disable interrupt: processor will ignore further interrupts whilst processing one interrupt.

Define priorities: low priority interrupts can be interrupted by higher priority interrupts.

Sequential multiple interrupts



Nested multiple interrupt.



System bus: A bus that connects major computer components (processor, memory, I/O) is called system bus.

Single bus problem:

1. propagation delay
2. If aggregate data transfer approach bus capacity.

multiple buses overcome these problem.

Timing Diagram

1. Synchronous.
2. Asynchronous.

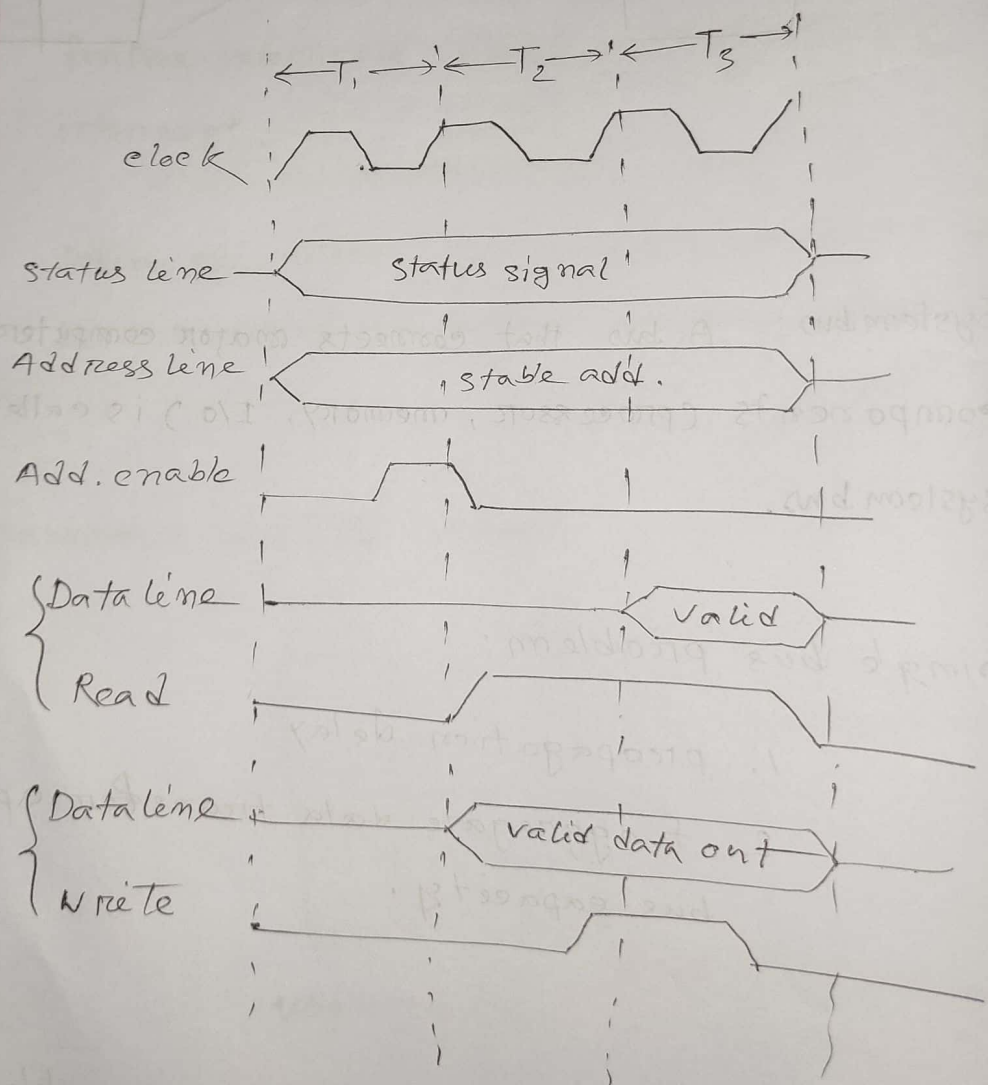
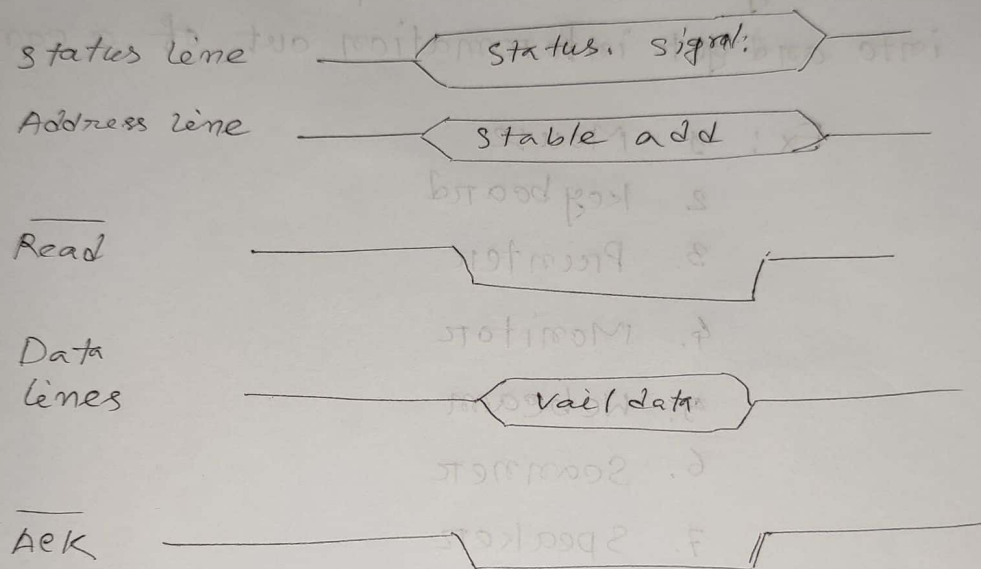


Fig: Synchronous

Asynchronous timing - Read diagram



PCI bus

1. peripheral components interconnection
2. Intel released to public domain
3. 32 or 64 bit
4. 50 lines.

peripheral Device: A peripheral device is an auxiliary device used to put information into and get information out of a computer.

- Ex:
1. Mouse
 2. Keyboard
 3. Printer
 4. Monitor
 5. Webcam
 6. Scanner
 7. Speaker

Computer architecture and organization

CH-03

Topic: Bus [set of channel] ✓

Bus: A Bus is a communication pathway connecting two or more devices. A key characteristics of a bus is that it is shared transmission medium.

System Bus:

A Bus that connects major computer components (processor, memory, I/O) is called system bus.

Types of buses:

1. Address Bus
2. Data Bus
3. Control Bus (timing information, signal)

☐ Data Bus: Data ^{line} Bus provide a path for moving data between system module. These lines, collectively are called data bus.

- ✓ 32-100 separate line
- ✓ Bidirectional.
- ✓ If the data bus is 8 bit wide and each instruction is 16 bits long, processor access the memory module twice

Address line: The address line are used to designate the source or destination of the data on the data bus.

- ✓ determines the max memory

- ✓ unidirectional

Control line: The control lines are used to control the access to and the use of data and address line

- ✓ transmitted both command and information.

- ✓ Bidirectional.

□ Control line include the following

1. Memory Write:

2. Memory Write Read:

3. I/O write

4. I/O read

5. Transfer ACK

6. Bus request.

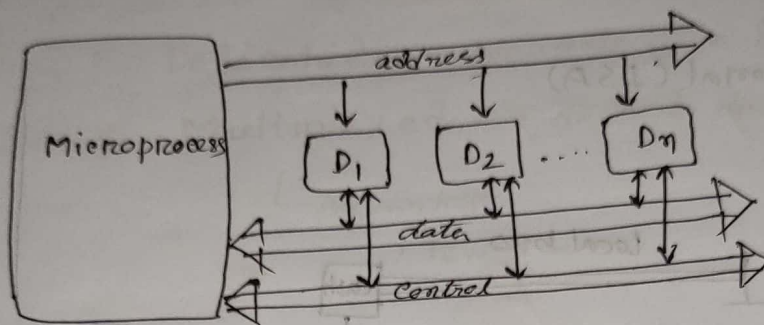
7. Bus grant

8. Interrupt request

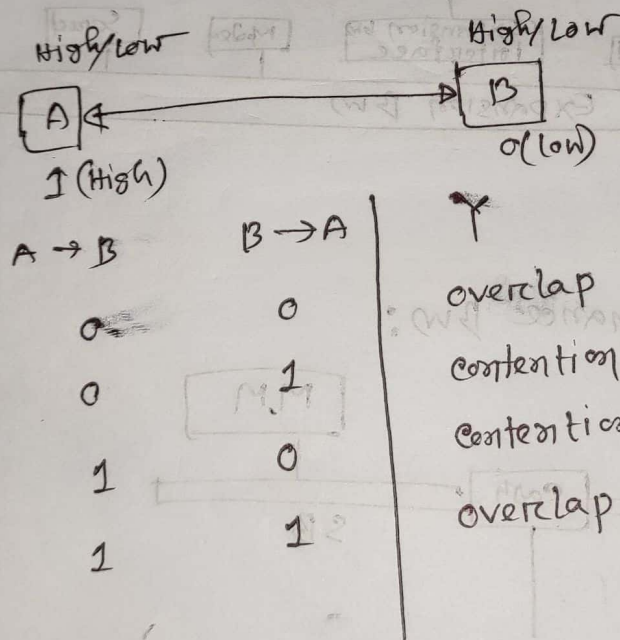
9. Interrupt ACK

10. clock

11. Reset.



Problem

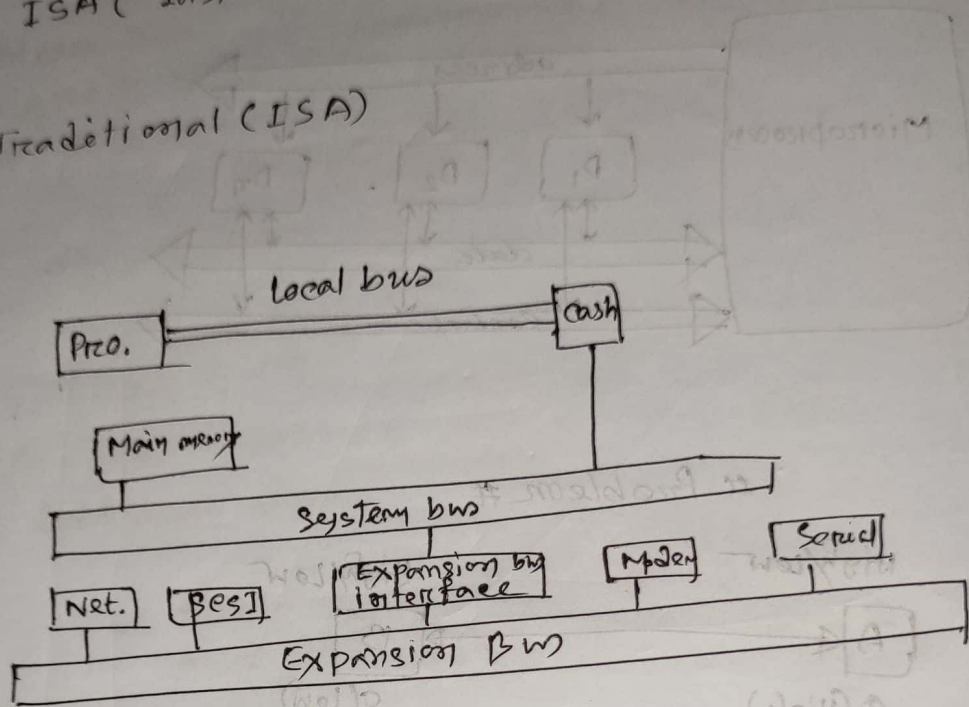


Remedy

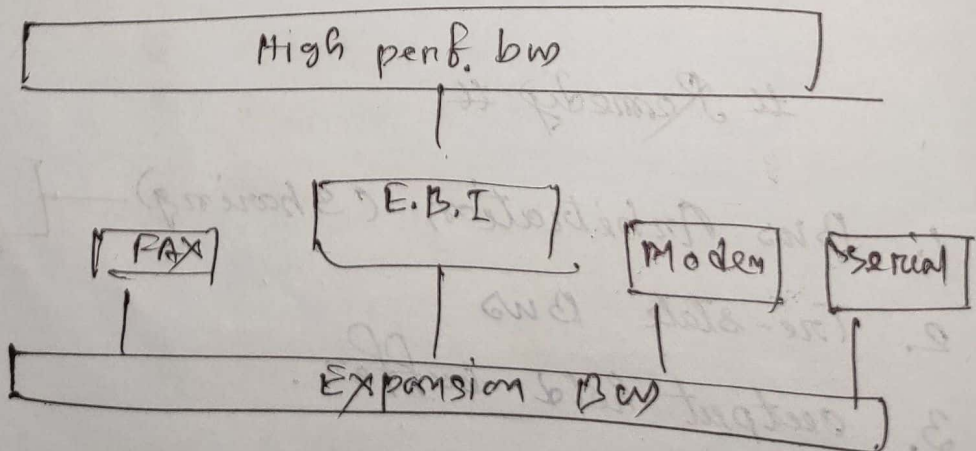
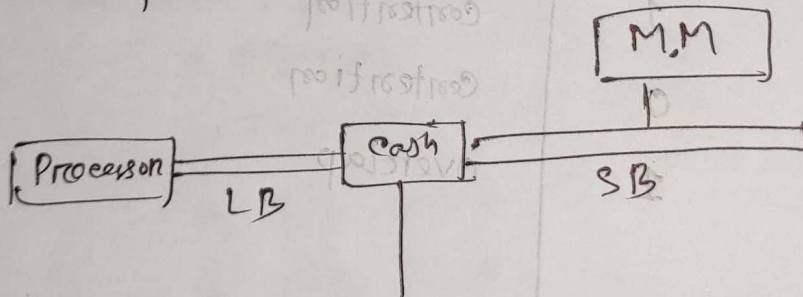
1. Bus Arbitration (Sharing) — { centralized (DM)
decentralized
2. Tri-state Bus
3. output hold buffer.

ISA (Instruction Set Architecture)

Traditional (ISA)



High performance BW:



There are two types of bus

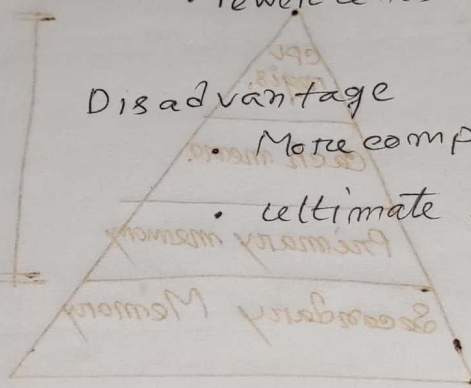
1. Dedicated $\rightarrow$ separate data & address line
2. Multiplexed $\rightarrow$ shared data and address line

Advantage

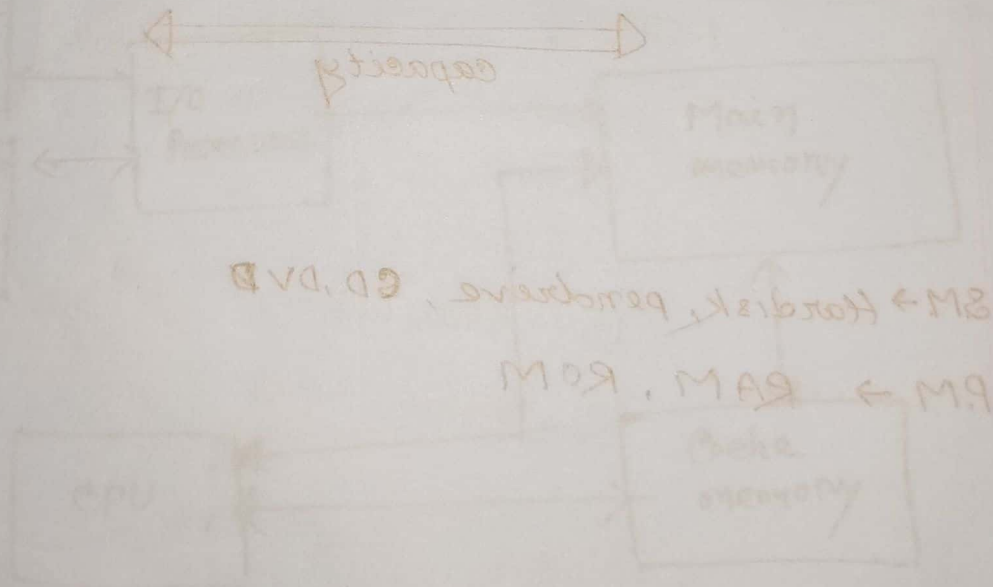
- Fewer lines

Disadvantage

- More complex control
- Ultimate performance



Bus



❏ peripheral Device: A peripheral device is an auxiliary device used to put information into and get information out of a computer.

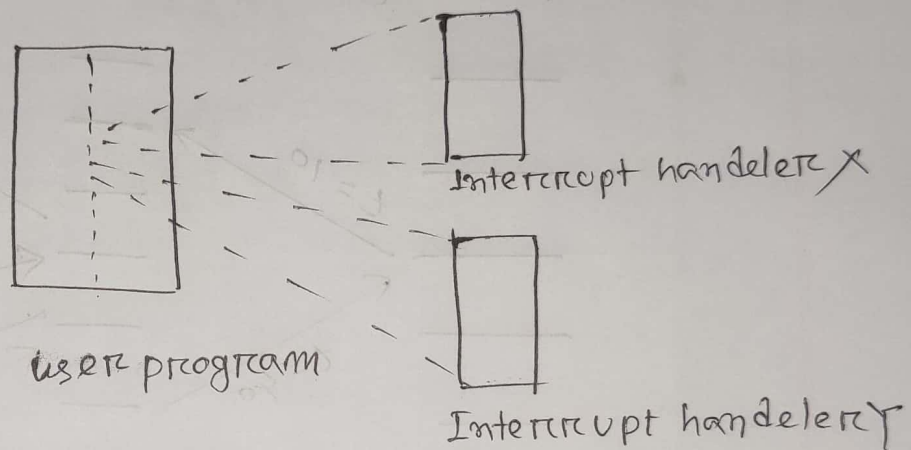
- Ex:
1. Mouse
 2. Keyboard
 3. Printer
 4. Monitor
 5. Webcam
 6. Scanner
 7. Speaker

❏ Characteristics of computer family.

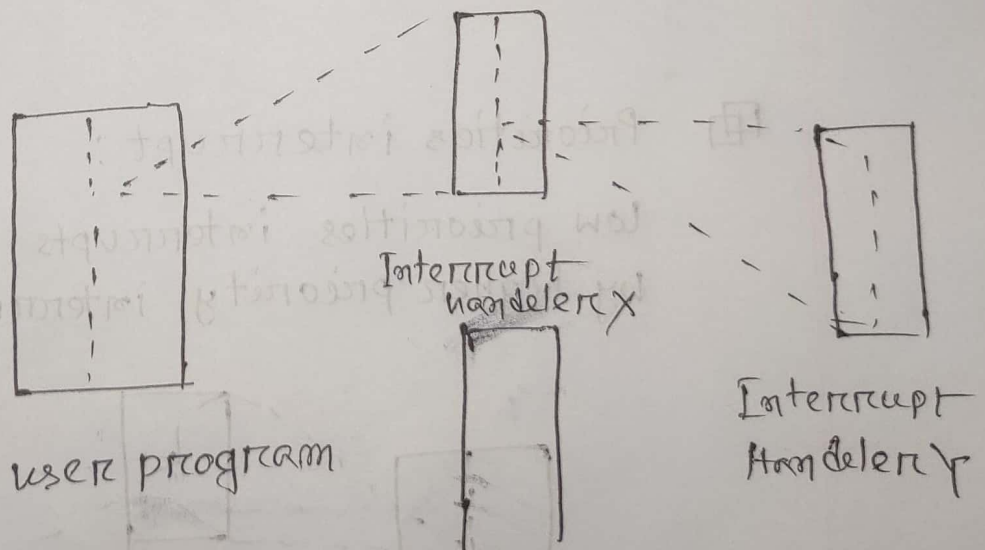
1. Similar or Identical set
2. Similar or Identical operating system
3. Increasing speed
4. Increasing Number of I/O ports
5. Increasing memory size
6. Increasing cost

Sequential interrupt:

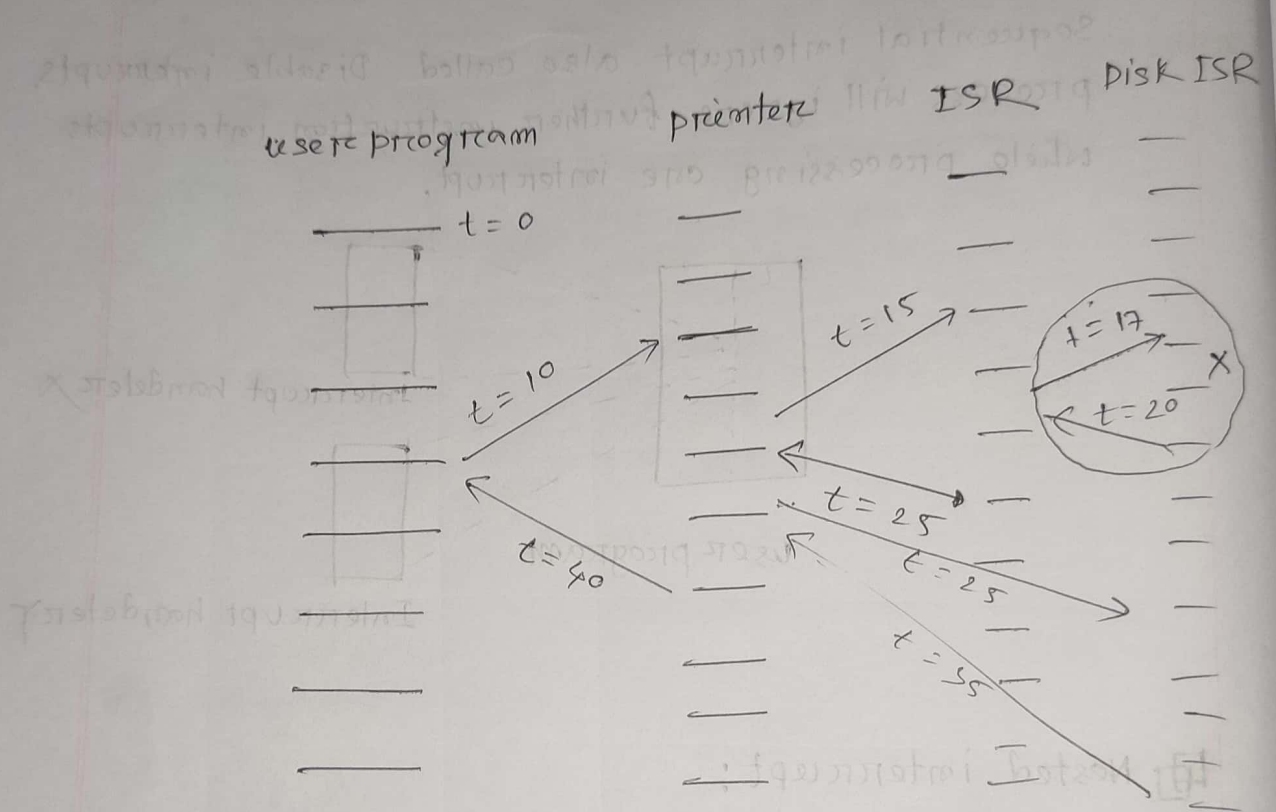
Sequential interrupt also called Disable interrupts, process will ignore further instruction interrupts while processing one interrupt.



Nested interrupt:

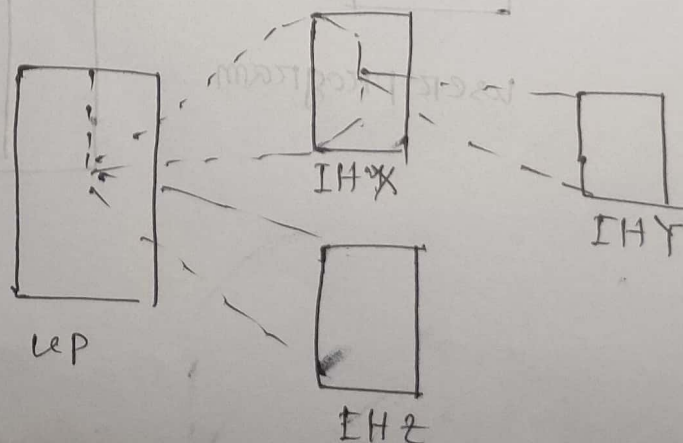


time sequence Interrupt



Priorities interrupt:

low priorities interrupts can be interrupted by higher priority interrupts



4 bit → 4004

4040

8 bit → 8080

8085

16 bit → 8086

80286

32 bit → 80386

80486

64 bit → Intel core 2

Intel core i7, to recent.

The IAS Instruction Set:

Instruction type	opcode	symbolic repres.	Description
Data transfer	0000 0001	LOAD M(x)	Transfer M(x) to the accumulator
Unconditional Branch	0000 1101	JUMP M(x, 0:19)	Take next instruction from left half of M(x)
Conditional Branch	0000 1111	JUMPT M(x, 0:19)	If number in the accumulator is nonnegative take next instruction from left half of M(x)
Arithmetic	0000 0101	ADD M(x)	Add M(x) to AC
Address modify	0001 0010	STOR M(x, 8:19)	replace left address field at M(x) by 12 rightmost bits of AC

▣ Basic Elements to Design Different buses

1. Bus type:

- Dedicated
- Multiplexed

2. Method of arbitration

- Centralized
- Decentralized

3. Timing

- Synchronous timing
- Asynchronous timing

4. Bus width

5. Data transfer type

6. Block data transfer